

Project 2: Automated AI Summaries with Agentic AI: n8n, Lambda, and Claude on Bedrock

Use Case: Summarizing S3 Uploads and Emailing a Note

Implement the following flow using n8n, Lambda and Bedrock:

1. A user uploads a file to an S3 bucket.
2. n8n detects the new file in S3.
3. n8n triggers an AWS Lambda function.
4. Lambda processes the file using Bedrock Claude to generate a summary.
5. n8n sends the summary via email.

Reference

Use the document n8n hands-on to go through a step by step demo of n8n workflow creation

You can also see several pre-built workflows at:

<https://blog.n8n.io/ai-agentic-workflows/>

Components:

- **AWS S3:** Your storage for uploaded files.
- **n8n:** The workflow automation platform that orchestrates the process.
- **AWS Lambda:** A serverless compute service to run the summarization logic.
- **AWS Bedrock:** Provides access to the Claude LLM for summarization.
- **Email Service (e.g., SendGrid, AWS SES, Gmail):** To send the summarized note.

Implementation Details

1. **S3 Trigger (n8n):**
 - Use the "Amazon S3" trigger node in n8n.
 - Configure it to watch for "Object Create" events in your specific S3 bucket.
 - This trigger will start the workflow whenever a new file is uploaded.
2. **Get S3 Object (n8n):**

- Use the "Amazon S3" node with the "Get Object" operation.
- Use the data from the S3 trigger (the file key/name) to download the uploaded file content.

3. **Invoke Lambda (n8n):**

- Use the "AWS Lambda" node.
- Configure it to invoke your Lambda function (see Lambda function details below).
- Pass the file content (from the "Get S3 Object" node) as input to the Lambda function.

4. **Lambda Function (Python):**

Implement a Lambda function that will do the following:

- **Get File Content:** The Lambda function receives the file content from n8n.
- **Invoke Bedrock Claude:** Uses the boto3 library to call the Bedrock API. You'll need to configure the appropriate model ID (anthropic.claude-v2 or similar) and any desired parameters (e.g., max_tokens_to_sample, temperature).
- **Extract Summary:** Parses the JSON response from Bedrock and extracts the generated summary.
- **Return Summary:** The Lambda function returns the summary as a JSON object.
- **Illustrative code:**

```
import boto3
```

```
import json
```

```
bedrock = boto3.client(service_name='bedrock-runtime')
```

```
def lambda_handler(event, context):
```

```
    file_content = event['file_content'] # Get file content from n8n
```

```
    # Choose your Bedrock Claude model
```

```

model_id = 'anthropic.claude-v2'
accept = 'application/json'
content_type = 'application/json'

prompt = f"Summarize the following text:\n\n{file_content}"

body = json.dumps({
    "prompt": prompt,
    "max_tokens_to_sample": 500, # Adjust as needed
    # Add other Claude parameters as required (temperature, etc.)
})

response = bedrock.invoke_model(
    body=body, modelId=model_id, accept=accept,
    contentType=content_type
)

response_body = json.loads(response['body'].read())
summary = response_body['completion']

return {
    'summary': summary
}

```

5. **Get Lambda Response (n8n):**

- The "AWS Lambda" node in n8n receives the JSON response from the Lambda function.

6. **Email Note (n8n):**

- Use an email node (e.g., "Gmail," "SendGrid," or "SMTP") to send the summarized note.

- Configure the "To," "From," "Subject," and "Body" fields.
- In the email body, use an expression to include the summary from the Lambda response: `{{ $json.summary }}`.

Important Considerations

- **Permissions:** Ensure that:
 - The Lambda function has permissions to access Bedrock.
 - The n8n instance has credentials to access S3, Lambda, and your email service.
- **Error Handling:** Add error-handling branches in your n8n workflow to gracefully handle potential issues (e.g., file download failures, Lambda invocation errors, Bedrock API errors).
- **File Size Limits:** Be mindful of file size limits for both Lambda and Bedrock. You may need to implement chunking or alternative strategies for very large files.
- **Cost:** Using Lambda and Bedrock incurs costs based on usage. Monitor your AWS costs carefully.
- **Security:** Securely store your AWS credentials and API keys.

Submit

In the shared folder, place the following:

- 1) Presentation PPT
- 2) Working code components with any steps/aws setup documented in the code